

GIRRAWEE HIGH SCHOOL

MATHEMATICS

Year 11

April, 2010

Task 2 Part A

Time Allowed: 45 minutes

INSTRUCTIONS:

- Attempt all questions.
- Write your answers on your own paper.
- All necessary working must be shown.
- Marks may be deducted for careless or badly arranged work.

Question 1 (20 marks)

- a) Find, correct to 4 significant places: 1

$$\frac{3.24^2}{5.73 - 2.84}$$

- b) If $\sqrt{5x} = 5\sqrt{2}$, find the value of x . 2

- c) Express 0.00205 in scientific notation. 1

- d) Simplify, giving exact answers:

(i) $\sqrt{20} + \sqrt{125} - \sqrt{45}$ (ii) $2\sqrt{3}(4\sqrt{2} - 5\sqrt{3})$ (iii) $(\sqrt{5} + 2\sqrt{3})^2$ 6

- e) Rationalise the denominator: $\frac{\sqrt{6}}{3\sqrt{6} - \sqrt{2}}$ 3

- f) Find the values of p and q if $\frac{\sqrt{5}}{\sqrt{5} - 2} = p + q\sqrt{5}$ 2

- g) Find the exact value of $\frac{A^2 B^3}{C^2}$ where $A = \left(\frac{5}{2}\right)^2$, $B = \left(\frac{2}{3}\right)^3$, $C = \left(\frac{5}{3}\right)^3$ 3

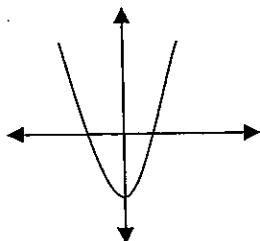
- h) Express $0.\dot{2}3\dot{5}$ as a fraction in its simplest form. 2

Question 2 (22 marks)

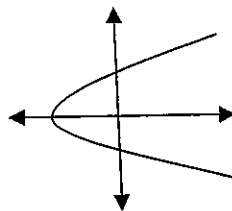
a) Indicate whether the following curves are functions or not:

4

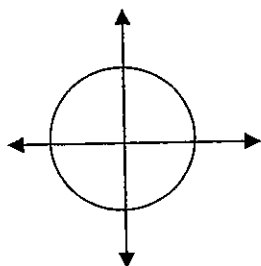
(i)



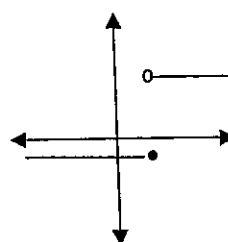
(ii)



(iii)



(iv)



b) For each of the following:

16

- (i) Sketch the curve
- (ii) State whether they are functions
- (iii) State the domain and range

(α) $x^2 + y^2 = 9$

(β) $y = 4^x$

(γ) $y = |x - 1|$

(δ) $y = \frac{1}{x+1}$

c) State the domain of each function:

2

(i) $y = \sqrt{5-x}$

(ii) $y = \frac{1}{\sqrt{16-x^2}}$

Question 3 (34 marks)

a) Find the exact value of:

6

(i) $\cos 135^\circ$

(ii) $\tan 330^\circ$

(iii) $\sin (-150^\circ)$

b) Given that $\cos \theta = \frac{2}{5}$ and $\sin \theta < 0$, find the exact value of $\tan \theta$.

2

c) Simplify:

4

(i) $\frac{\sin x}{\tan x}$

(ii) $\frac{1}{\cos x} - \sin x \tan x$

d) Prove the following identities:

9

(i) $3 - 4\cos^2 x = 4\sin^2 x - 1$

(ii) $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 = 2$

(iii) $\frac{\cos x}{1 - \sin x} - \frac{\cos x}{1 + \sin x} = 2 \tan x$

e) Sketch $y = \cos \theta$, $0^\circ \leq \theta \leq 360^\circ$.

2

f) Solve for $0^\circ \leq \theta \leq 360^\circ$

11

(i) $2\cos \theta = \sqrt{3}$

(ii) $\operatorname{cosec}^2 \theta - 2 = 0$

(iii) $\tan 2\theta = \frac{1}{\sqrt{3}}$

(iv) $\sin^2 \theta + \sin \theta = 0$

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MATHEMATICS

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Task 2 Part B

Time Allowed: 45 minutes

INSTRUCTIONS:

- Attempt all questions.
- Write your answers on your own paper.
- All necessary working must be shown.
- Marks may be deducted for careless or badly arranged work.

Question 4 (31 marks)

a) Factorise:

- | | | |
|-------|------------------|---|
| (i) | $16 - 25x^2$ | 1 |
| (ii) | $3x^2 + 14x - 5$ | 2 |
| (iii) | $x^3 - 64$ | 2 |

b) Simplify:

- | | | |
|-------|--|---|
| (i) | $\frac{a^4x^2 - b^4x^2}{ax + bx}$ | 3 |
| (ii) | $\frac{3}{x+6} - \frac{1}{x+2}$ | 3 |
| (iii) | $\frac{6x^2 + x - 2}{3x^2 - 3} \times \frac{4x^3 + 4x^2 - x - 1}{6x^2 + 7x + 2}$ | 3 |

c) Solve:

- | | | |
|-------|--|---|
| (i) | $\frac{2x-1}{3} = 1 - \frac{x-2}{5}$ | 3 |
| (ii) | $2x - 3 < 5$ | 2 |
| (iii) | $4 - 3x \geq 2 - 2x$ | 2 |
| (iv) | $(x+1)(x-3) > 0$ | 2 |
| (v) | $ x+3 = 5$ | 2 |
| (vi) | $\left \frac{3x+2}{5} \right \leq 4$ | 3 |
| (vii) | $ 2m-5 > 9$ | 3 |

Question 5 (11 marks)

- a) A function is defined by the rule:

$$f(x) = \begin{cases} -2 & \text{if } x \leq -5 \\ 0 & \text{if } -5 < x < 2 \\ x & \text{if } x \geq 2 \end{cases}$$

Find $f(-6) + f(1) + f(6)$

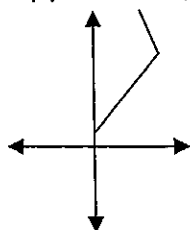
2

- b) Find the coordinates of the centre and the length of the radius of the circle

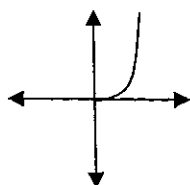
$$x^2 + y^2 + 6x - 10y + 18 = 0$$

4

- c) Copy and complete this diagram to give an even function:

1

- d) Copy and complete this diagram to give an odd function:

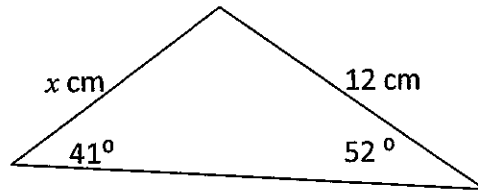
1

- e) Determine whether the function $f(x) = 2x^4 + 3x$ is even or odd or neither. Show all your working.

3

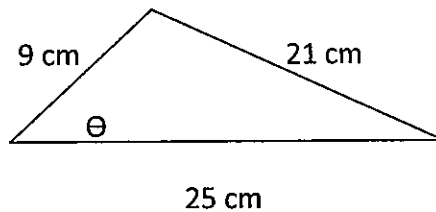
Question 6 (17 marks)

- a) The angle of depression of a car, from the top of a 300m tower is $59^{\circ}02'$. 2
How far from the tower is the car?
- b) A bushwalker trekked west from town A for 40km and then 75km south. 3
Find the bearing of the bushwalker from town A.
- c)



- (i) Find the value of x , correct to 2 decimal places. 2
(ii) Find the area of the triangle. 2

d)



Find the value of θ , correct to the nearest minute. 2

- e) Sam drove from his home for 200 km on a bearing of 040° , then drove on a bearing of 157° for 345 km.
- (i) Draw a diagram showing this information. 1
(ii) How far from home is Sam? 2
(iii) What is his bearing from home? 3

Solutions

Question 1 (20 marks)

a) 3.632 (1)

b) $\sqrt{5x} = 5\sqrt{2}$
 $= \sqrt{25 \times 2}$
 $= \sqrt{50}$
 $\therefore 5x = 50$
 $x = 10$ (2)

c) $0.00205 = 2.05 \times 10^{-3}$ (1)

d) i) $\sqrt{20} + \sqrt{125} - \sqrt{45}$
 $= 2\sqrt{5} + 5\sqrt{5} - 3\sqrt{5}$
 $= 4\sqrt{5}$ (2)

ii) $2\sqrt{3}(4\sqrt{2} - 5\sqrt{3})$
 $= 8\sqrt{6} - 30$ (2)

iii) $(\sqrt{5} + 2\sqrt{3})^2$
 $= 5 + 4\sqrt{15} + 12$
 $= 17 + 4\sqrt{15}$ (2)

e) $\frac{\sqrt{6}}{3\sqrt{6} - \sqrt{2}} \times \frac{3\sqrt{6} + \sqrt{2}}{3\sqrt{6} + \sqrt{2}}$
 $= \frac{18 + \sqrt{12}}{52}$
 $= \frac{18 + 2\sqrt{3}}{52}$
 $= \frac{9 + \sqrt{3}}{26}$ (3)

f) $\frac{\sqrt{5}}{\sqrt{5} - 2} \times \frac{\sqrt{5} + 2}{\sqrt{5} + 2}$
 $= \frac{5 + 2\sqrt{5}}{5 - 4}$
 $= 5 + 2\sqrt{5} = p + q\sqrt{5}$
 $\therefore p = 5, q = 2$ (2)

g) $\frac{A^2 B^3}{C^2}$
 $= \frac{\left[\left(\frac{5}{2}\right)^2\right]^2 \left[\left(\frac{2}{3}\right)^3\right]^3}{\left[\left(\frac{5}{3}\right)^3\right]^2}$
 $= \frac{5^4}{2^4} \times \frac{2^9}{3^9} \times \frac{3^6}{5^6}$
 $= \frac{2^5}{5^2 \times 3^3}$
 $= \frac{32}{675}$ (3)

h) $0.\dot{2}3\dot{5}$
 $x = 0.235235 \dots$
 $1000x = 235.235 \dots$
 $999x = 235$
 $x = \frac{235}{999}$ (2)

Question 2 (22 marks)

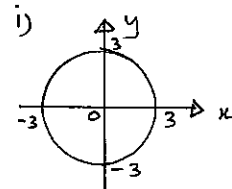
a) i) function

ii) not a function

iii) not a function

iv) function. (4)

b) i) $x^2 + y^2 = 9$

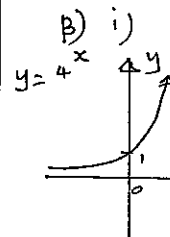


ii) Not a function

iii) $D: -3 \leq x \leq 3$

$R: -3 \leq y \leq 3$

(4)



B) i)

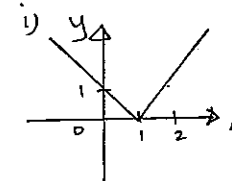
ii) Function

iii) $D: \text{all real } x$

$R: y > 0$

(4)

c) $y = |x - 1|$



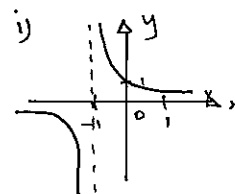
ii) Function

iii) $D: \text{all real } x$

$R: y \geq 0$

(4)

d) $y = \frac{1}{x+1}$



ii) Function

iii) $D: \text{all real } x, x \neq -1$

$R: \text{all real } y, y \neq 0$

(4)

c) $y = \sqrt{5 - x}$

$D: x \leq 5$ (1)

d) $y = \frac{1}{\sqrt{16 - x^2}}$

$D: -4 < x < 4$ (1)

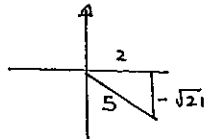
Question 3 (34 marks)

i) $\cos 135^\circ$ (Q2)
 $= \cos (180^\circ - 45^\circ)$
 $= -\cos 45^\circ$
 $= -\frac{1}{\sqrt{2}}$ (2)

ii) $\tan 330^\circ$ (Q4)
 $= \tan (360^\circ - 30^\circ)$
 $= -\tan 30^\circ$
 $= -\frac{1}{\sqrt{3}}$ (2)

iii) $\sin (-150^\circ)$ (Q3)
 $= \sin 210^\circ$
 $= \sin (180^\circ + 30^\circ)$
 $= -\sin 30^\circ$
 $= -\frac{1}{2}$ (2)

b) $\cos \theta = \frac{2}{5}$, $\sin \theta < 0$ [Q4]



$$\tan \theta = -\frac{\sqrt{21}}{2}$$

(2)

c) i) $\frac{\sin x}{\tan x}$
 $= \sin x \times \frac{\cos x}{\sin x}$
 $= \cos x$ (2)

ii) $\frac{1}{\cos x} - \sin x \tan x$
 $= \frac{1}{\cos x} - \sin x \cdot \frac{\sin x}{\cos x}$
 $= \frac{1}{\cos x} - \frac{\sin^2 x}{\cos x}$
 $= \frac{1 - \sin^2 x}{\cos x}$
 $= \frac{\cos^2 x}{\cos x}$
 $= \cos x$ (2)

d) i) $3 - 4\cos^2 u - \sin^2 u - 1$

$$\begin{aligned} \text{LHS} &= 3 - 4\cos^2 u \\ &= 3 - 4(1 - \sin^2 u) \\ &= 3 - 4 + 4\sin^2 u \\ &= 4\sin^2 u - 1 = \text{RHS} \end{aligned}$$
 (2)

ii) $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 = 2$

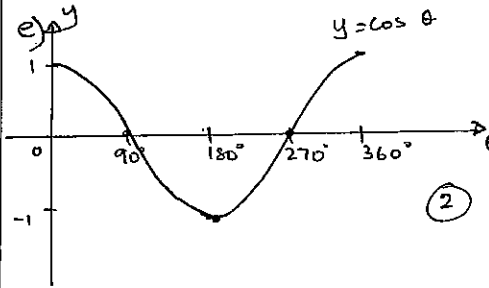
$$\begin{aligned} \text{LHS} &= (\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 \\ &= \sin^2 \theta + 2\sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta - 2\sin \theta \cos \theta + \cos^2 \theta \\ &= 2\sin^2 \theta + 2\cos^2 \theta \\ &= 2(\sin^2 \theta + \cos^2 \theta) \\ &= 2 = \text{RHS} \end{aligned}$$
 (3)

iii) $\frac{\cos x}{1 - \sin x} - \frac{\cos x}{1 + \sin x} = 2 \tan x$

$$\begin{aligned} \text{LHS} &= \frac{\cos x}{1 - \sin x} - \frac{\cos x}{1 + \sin x} \\ &= \frac{\cos x(1 + \sin x) - \cos x(1 - \sin x)}{(1 - \sin x)(1 + \sin x)} \\ &= \frac{\cos x + \cos x \sin x - \cos x + \cos x \sin x}{(1 - \sin x)(1 + \sin x)} \\ &= \frac{2 \cos x \sin x}{1 - \sin^2 x} \\ &= \frac{2 \sin x \cos x}{\cos^2 x} \\ &= \frac{2 \sin x}{\cos x} \end{aligned}$$

$$= 2 \tan x = \text{RHS}$$
 (3)

Question 3 (Cont.)



(2)

f) i) $2 \cos \theta = \sqrt{3}$

$$\cos \theta = \frac{\sqrt{3}}{2} \quad \text{Q1, 4}$$

$$[\cos 30^\circ = \frac{\sqrt{3}}{2}]$$

$$\begin{aligned} \theta &= 30^\circ, (360^\circ - 30^\circ) \\ &= 30^\circ, 330^\circ \end{aligned}$$
 (2)

ii) $\csc^2 \theta - 2 = 0$

$$\csc^2 \theta = 2$$

$$\frac{1}{\sin^2 \theta} = 2$$

$$\sin^2 \theta = \frac{1}{2}$$

$$\sin \theta = \pm \frac{1}{\sqrt{2}} \quad \text{Q1, 2, 3, 4}$$

$$[\sin 45^\circ = \frac{1}{\sqrt{2}}]$$

$$\theta = 45^\circ, (180^\circ - 45^\circ), (180^\circ + 45^\circ), (360^\circ - 45^\circ)$$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

(3)

iii) $\tan 2\theta = \frac{1}{\sqrt{3}} \quad \text{Q1, 3} \quad 0^\circ \leq \theta \leq 360^\circ$
 $0^\circ \leq 2\theta \leq 720^\circ$

$$[\tan 30^\circ = \frac{1}{\sqrt{3}}]$$

$$\begin{aligned} 2\theta &= 30^\circ, 210^\circ, 390^\circ, 570^\circ \\ \theta &= 15^\circ, 105^\circ, 195^\circ, 285^\circ \end{aligned}$$
 (3)

iv) $\sin^2 \theta + \sin \theta = 0$

$$\sin \theta (\sin \theta + 1) = 0$$

$$\sin \theta = 0 \quad \text{or} \quad \sin \theta = -1$$

$$\theta = 0^\circ, 180^\circ, 360^\circ \quad \theta = 270^\circ$$

$$\theta = 0^\circ, 180^\circ, 270^\circ, 360^\circ$$
 (3)

Question 4 (31 marks)

a) i) $16 - 25x^2$

$$= (4 - 5x)(4 + 5x) \quad (1)$$

ii) $3x^2 + 14x - 5$

$$\begin{matrix} x-15 \\ +14 \end{matrix}$$

$$= 3x^2 + 15x - x - 5$$

$$= 3x(x+5) - 1(x+5)$$

$$= (3x-1)(x+5) \quad (2)$$

iii) $x^3 - 64 = x^3 - 4^3$

$$= (x-4)(x^2 + 4x + 16) \quad (2)$$

b) i) $\frac{a^4x^2 - b^4x^2}{ax + bx}$

$$= \frac{x^2(a^4 - b^4)}{x(a+b)}$$

$$= \frac{x(a^2+b^2)(a^2-b^2)}{(a+b)}$$

$$= \frac{x(a^2+b^2)(a+b)(a-b)}{(a+b)}$$

$$= x(a-b)(a^2+b^2) \quad (3)$$

ii) $\frac{3}{x+6} - \frac{1}{2+2}$

$$= \frac{3(x+2) - (x+6)}{(x+6)(x+2)}$$

$$= \frac{3x+6 - x-6}{(x+6)(x+2)}$$

$$= \frac{2x}{(x+6)(x+2)} \quad (3)$$

iii) $\frac{6x^2+x-2}{3x^2-3} \times \frac{4x^3+4x^2-x-1}{6x^2+7x+2}$

$$= \frac{(2x-1)(3x+2)}{3(x+1)(x-1)} \times \frac{(4x^2-1)(x+1)}{(2x+1)(3x+2)}$$

$$= \frac{(2x-1)(2x-1)(2x+1)}{3(x-1)(2x+1)}$$

$$= \frac{(2x-1)^2}{3(x-1)} \quad (3)$$

c) i) $\frac{2x-1}{3} = 1 - \frac{x-2}{5} \quad (x15)$

$$5(2x-1) = 15 - 3(x-2)$$

$$10x - 5 = 15 - 3x + 6$$

$$13x = 26 \quad (3)$$

$$x = 2$$

ii) $2x - 3 < 5$

$$2x < 8$$

$$x < 4 \quad (2)$$

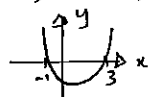
iii) $4 - 3x \geq 2 - 2x$

$$-x \geq -2$$

$$x \leq 2 \quad (2)$$

iv) $(x+1)(x-3) > 0$

$$x < -1, x > 3 \quad (2)$$



v) $|x+3| = 5$

$$x+3 = 5 \text{ or } x+3 = -5$$

$$x = 2, -8 \quad (2)$$

vi) $\left| \frac{3x+2}{5} \right| \leq 4$

$$-4 \leq \frac{3x+2}{5} \leq 4$$

$$-20 \leq 3x+2 \leq 20$$

$$-22 \leq 3x \leq 18$$

$$-\frac{22}{3} \leq x \leq 6 \quad (3)$$

vii) $|2m-5| > 9$

$$2m-5 > 9 \text{ or } 2m-5 < -9$$

$$2m > 14$$

$$2m < -4$$

$$m > 7$$

$$m < -2$$

$$m < -2, m > 7 \quad (3)$$

Question 5 (10 marks)

a) $f(-6) + f(1) + f(6)$

$$= -2 + 0 + 6$$

$$= 4 \quad (2)$$

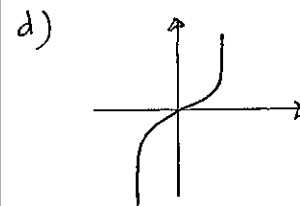
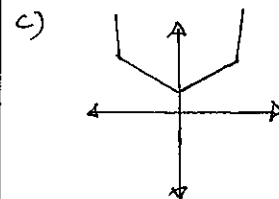
b) $x^2 + y^2 + 6x - 10y + 18 = 0$

$$x^2 + 6x + 9 + y^2 - 10y + 25 = -18 + 9 + 25$$

$$(x+3)^2 + (y-5)^2 = 16$$

$$\text{Centre} = (-3, 5)$$

$$\text{radius} = 4 \text{ units} \quad (4)$$



e) $f(x) = 2x^4 + 3x$

$$f(-x) = 2(-x)^4 + 3(-x)$$

$$= 2x^4 - 3x$$

$$f(-x) \neq f(x) \therefore \text{not even}$$

$$-f(x) = -2x^4 - 3x$$

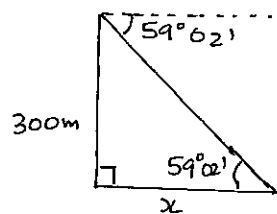
$$f(-x) \neq -f(x) \therefore \text{not odd}$$

$$\therefore f(x) = 2x^4 + 3x \text{ is}$$

$$\text{neither even nor odd.} \quad (3)$$

Question 6 (17 marks)

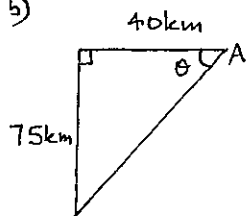
a)



$$\tan 59^{\circ}02' = \frac{300}{x}$$

$$x = \frac{300}{\tan 59^{\circ}02'} = 180\text{m} \quad (2)$$

b)



$$\tan \theta = \frac{75}{40}$$

$$\theta = 61^{\circ}56'$$

$$\text{Bearing} = 270^{\circ} - 61^{\circ}56' = 208^{\circ}04' \quad (3)$$

$$\Rightarrow \frac{x}{\sin 52^{\circ}} = \frac{12}{\sin 41^{\circ}}$$

$$x = \frac{12 \sin 52^{\circ}}{\sin 41^{\circ}} = 14.41\text{cm} \quad (2)$$

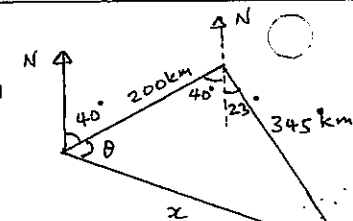
$$\begin{aligned} \text{ii) } A &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 12 \times 14.41 \sin 87^{\circ} \\ &= 86.36\text{cm}^2 \end{aligned} \quad (2)$$

$$\text{1) } \cos \theta = \frac{9^2 + 25^2 - 21^2}{2 \times 9 \times 25}$$

$$= 0.58$$

$$\theta = 53^{\circ}55' \quad (2)$$

e)



$$\begin{aligned} \text{ii) } x^2 &= 200^2 + 345^2 - 2 \times 200 \times 345 \cos 63^{\circ} \\ x &= 310\text{km} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{iii) } \frac{\sin \theta}{345} &= \frac{\sin 63}{310} \\ \sin \theta &= \frac{345 \sin 63}{310} \end{aligned}$$

$$\theta = 82^{\circ}34'$$

$$\begin{aligned} \therefore \text{Bearing} &= 40^{\circ} + 82^{\circ}34' \\ &= 122^{\circ}34' \end{aligned} \quad (3)$$